

Technology #150: Implement additively manufactured crack-resistant material inserts at rail joints to prevent hydrogen degradation and fatigue fracture

Technology Domain: mechanical_structural

Mechanism: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Implement additively manufactured crack-resistant material inserts at rail joints to prevent hydrogen degradation and fatigue fracture

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Next Experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

Buyer Receives: Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

Buyer Must Build: Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

Transfer Posture: Sponsored Validation

Target Buyer: domain-appropriate technology buyer (to be identified by portfolio owner)

CONFIDENTIAL. See full engineering dossier for details.